JUNIA Grande école d'ingénieurs
HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable

Information

- MOOC Evaluation

⚠Contrary to what is written, you will **only be allowed one attempt** for this assessment.

Hello everyone,

As a reminder, you must take the INR's Responsible Digital Knowledge Certificate following the MOOC you have completed.
This continuous assessment lasts 40 minutes and counts for 15% of the "Responsible Digital" course grade.

You must complete this assessment between October 6th and 19th, whether or not you have MOOC hours scheduled during this period. Failure to complete this test will result in a grade of 0.

Please note that **the conditions for taking the exam are strict** and are all described in the "Connection Instructions" section at the bottom of this page: https://institutnr.org/certificat-de-connaissance-nr

Here is an extract from the conditions imposed by the INR, which administers the exam:

- You must bring ID
- Have a webcam
- A room to isolate yourself (you must be alone in a closed room and will have to show your surroundings via the webcam)
- No documents are allowed during the exam, and you are not allowed to use dual screens
- To obtain the certification, you will need a score of 72.5%, or 29 correct answers out of 40 questions.

We will also set the academic mark for JUNIA at 72.5%. 20/20 if you get 100%, etc.

For any problems or questions: yohan.giraudeau@junia.com

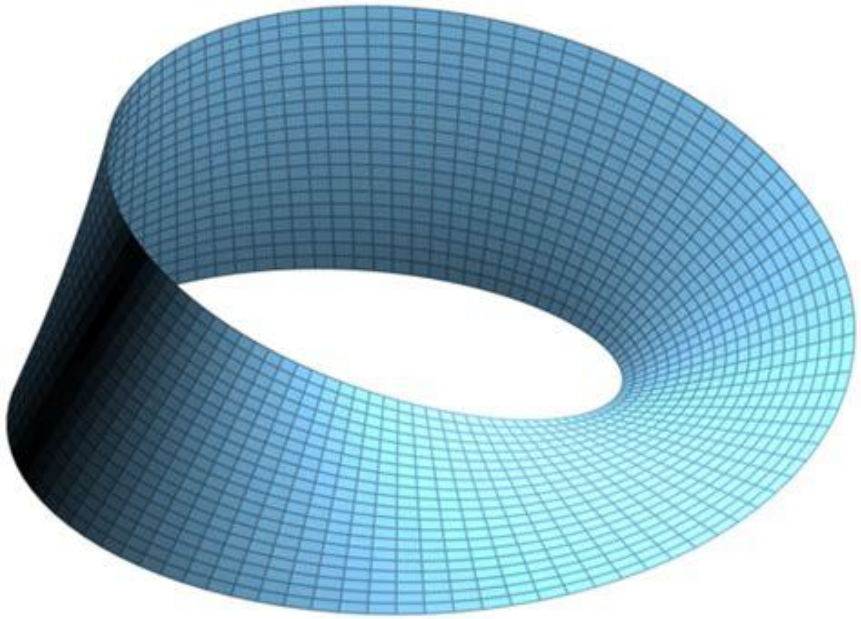
Lesson 4 : Environnemental Assessment | LCA

Lesson 4 - Today

- Introduction to Eco design
- Environmental assessment : LCA
- Practical application for a SI
- Best practices & regulation



What is Moebius ring ?







Which year did the first environmental impact analysis in industry take place?

1969

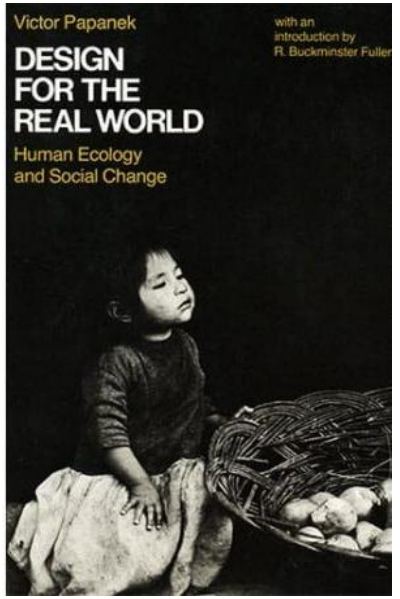
« the returnable glass bottle (provided it makes 15 trips before being thrown away) is the ecological container 'par excellence' »



Source : https://www.eco-conception.fr/static/h/histoire-de-l-eco-conception.html?n=20231126215639

"Le Design, s'il doit devenir écologiquement et socialement responsable soir être révolutionnaire et radical dans le vrai sens du terme. Cela signifie consommer moins, utiliser les choses plus longtemps et être économe en recyclant les matériaux. "

Victor PAPANEK, Austrian-American designer (1926-1998)



" Il faut créer une coopération spontanée sans dommages écologiques et inconvénients pour qui que ce soit afin que le monde fonctionne"

Richard Buckminster FULLER, american architect and designer (1895 - 1983)

What is ecodesign ?

Definition

- **Ecodesign** : Standard for reducing the environmental impact of a product or service.

- **ISO 14062 / IEC 62430:** *"Integration of environmental constraints into the design of products and services using a comprehensive, multi-criteria approach."*

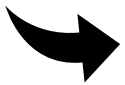
We cannot eco-design a software or a website

We eco-design a business process.

What about digital ?

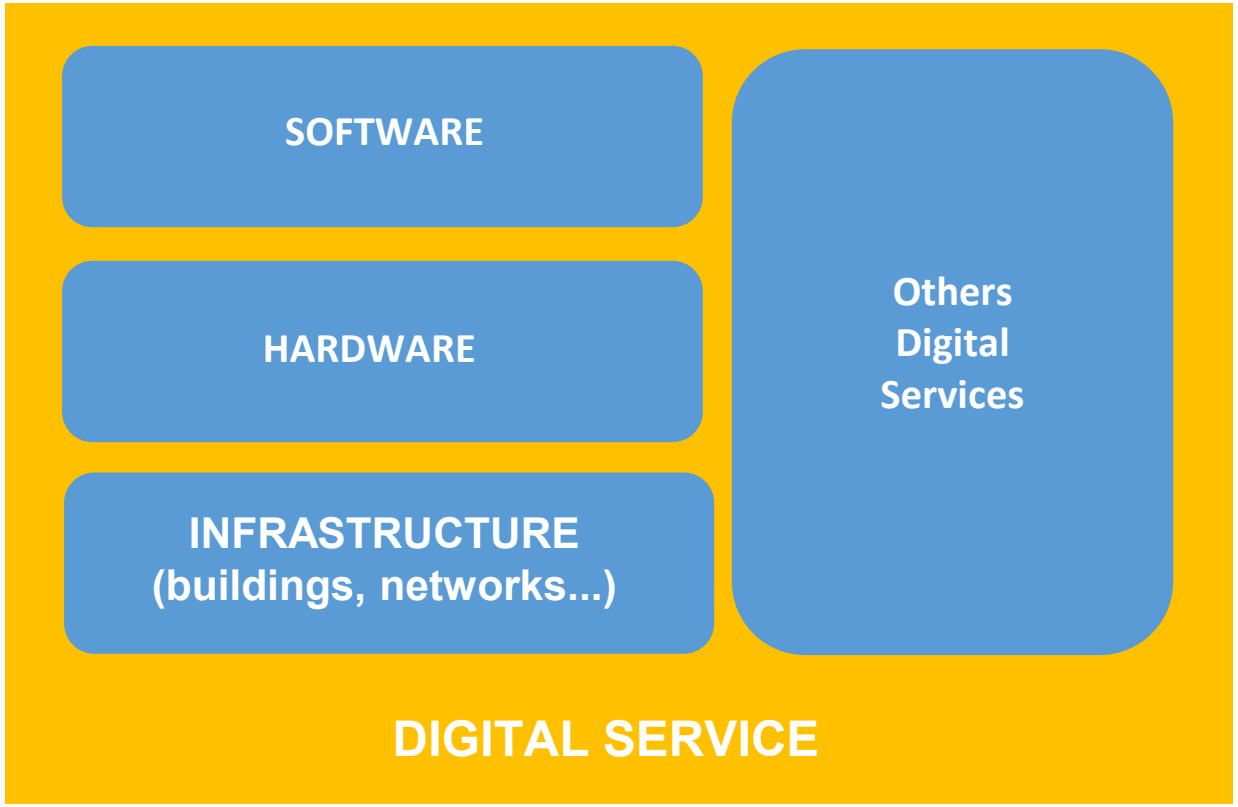
A lot of business process are done through digital :

- Booking a train ticket
- Buying a product
- Getting information by reading an article
- Finding a doctor's address
- Registering your children for school meals
- Taking a distance learning course



Behind every daily digital activity... There is a digital service !

A digital service

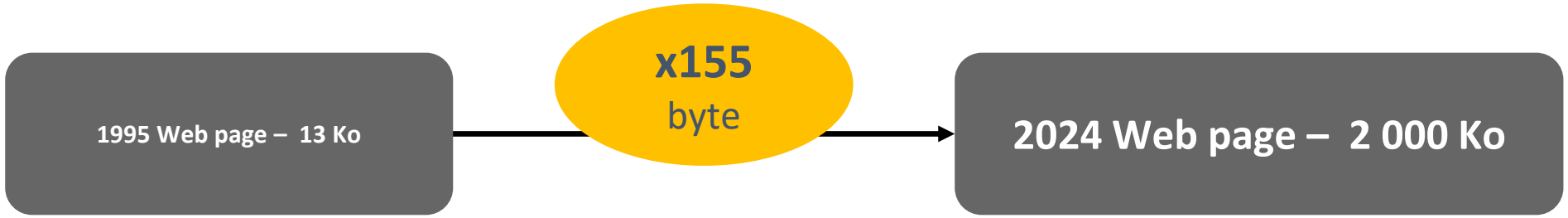


Source : Collectif Green IT, Du Green IT au numérique responsable, lexique, 2018



Why eco-design digital services?

It is urgent to slim down the software !



We don't replace our terminals because they no longer work... but because they're slow. They slow because **software gets fatter** over time.

sources : Ecoconception web : les 115 bonnes pratiques, Eyrolles, 2022, https://ecoconceptionweb.com/

Logiciel : la clé de l'obsolescence programmée du matériel informatique, GreenIT.fr, 2010-2022, https://www.greenit.fr/2010/05/24/logiciel-la-cle-de-l-obsolescence-programmee-du-materiel-informatique/

Ecodesign a digital service

When we ecodesign a digital service, we allow to :

- **Extend the lifespan** of equipment, particularly user terminals.
- **Reduce the amount of IT resources** required to operate the service.

The right balance

Most of digital weight *"fat digital"* is first business, functional and/or graphic fat.

Find the right balance : Minimum Valuable Product - MVP

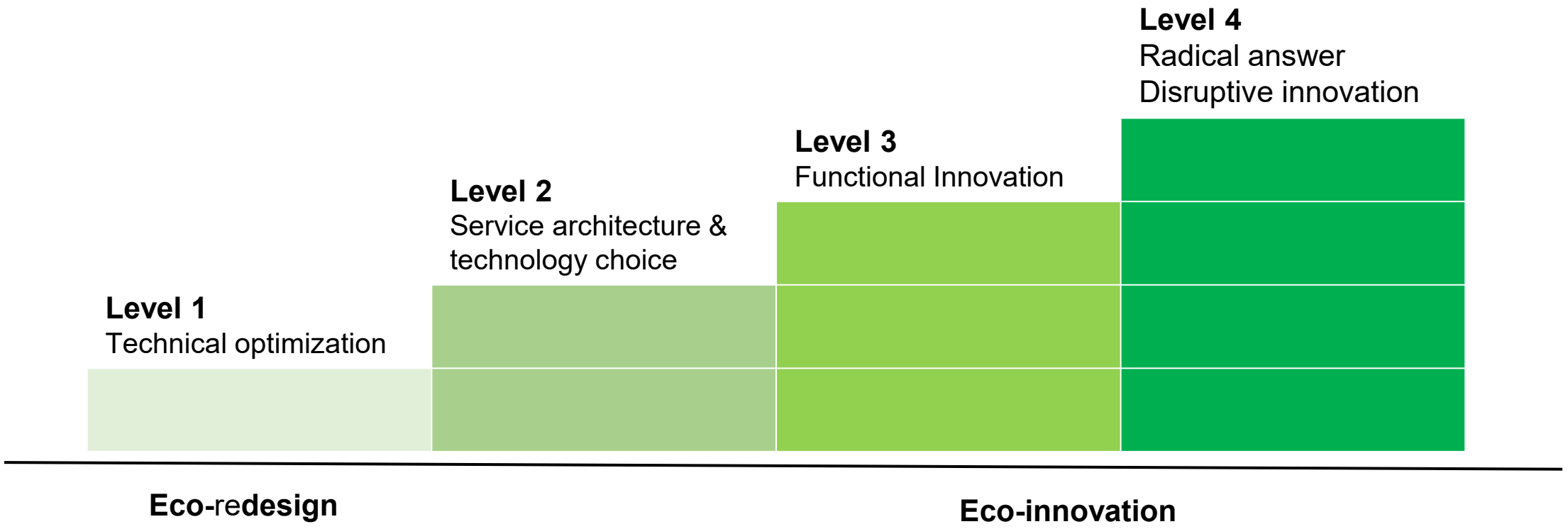


Up to 1160x more resources are required to deliver the same result (train schedule)!

From optimization to true eco-design

4 Levels of Ecodesign

> We're Looking for Radical Answers



How to evaluate ?

Definition / prerequisite

Environmental assessment

The purpose of environmental assessment is to **measure and analyze environmental impacts in order to prevent harmful consequences.**

Environmental assessments can be applied in a variety of ways. These include conducting an environmental assessment of :

- a public policy,
- a major project,
- a product (good or service)
- or an organization (e.g., a production site)

It helps identify foreseeable impacts, analyze and justify choices according to environmental challenges. It is a **decision-making tool.**



sources : Ademe, Ministère de l'environnement

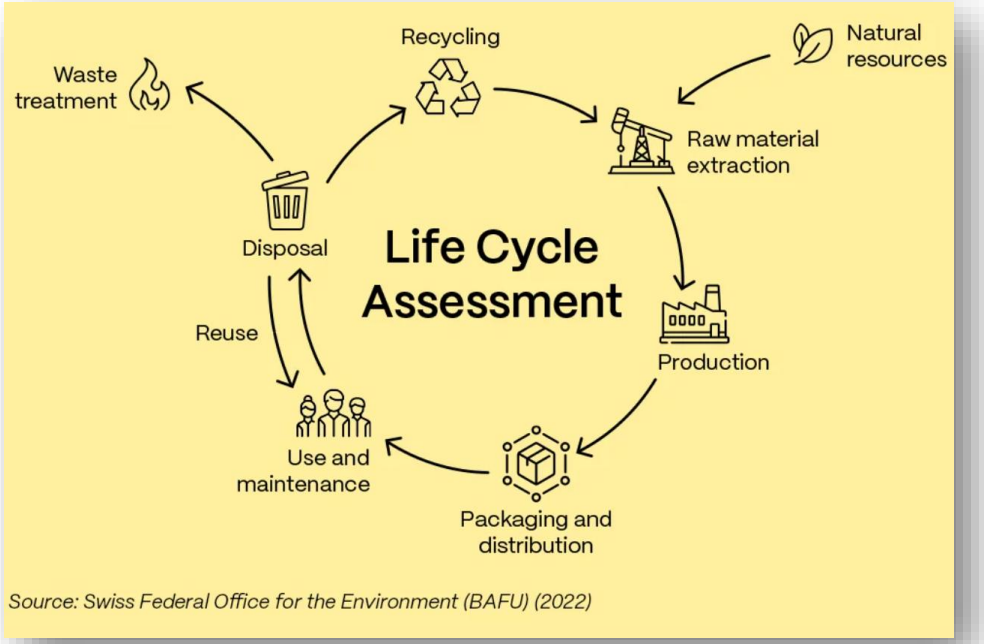
Definition / prerequisite

Life cycle

From environmental point of view, the life cycle of a product or service is the sequence of consecutive phases undergone by a product or a system of products, from the extraction of raw materials or the generation of natural resources to its final disposal.

The main phases are :

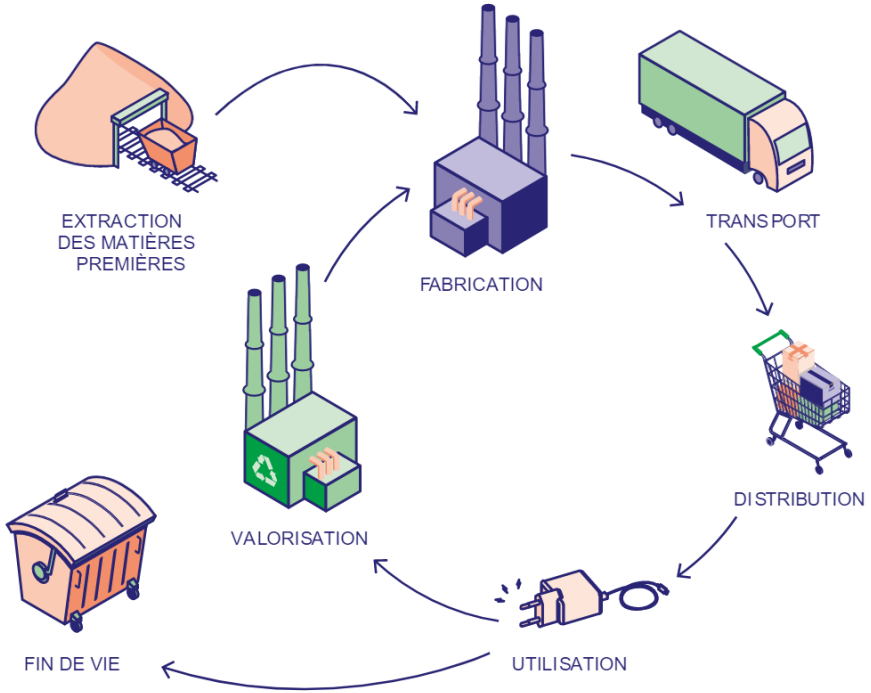
- **Manufacturing**, including the acquisition of raw materials and distribution (packaging and transportation)
- **Installation**
- **Use**
- **End of life** (recycling, reuse, recovery, incineration, landfill).



Source: Swiss Federal Office for the Environment (BAFU) (2022)

sources : Alliance Green IT / livre blanc ecoconception des services numeriques (2017), playhooky.fr

Life Cycle Assessment (LCA)



Method for assessing the environmental impacts of a product

Multi-criteria approach to environmental impacts: Incoming and outgoing material and energy flows at each stage of the life cycle are inventoried (life cycle inventory: LCI)

Consideration of the entire product life cycle "from cradle to grave" to avoid the transfer of pollution from one stage of the life cycle to another.

Standardized by **ISO: 14044/40**

Source : https://ecoresponsable.numerique.gouv.fr/

Life Cycle Analysis : what is it for ?

Life cycle assessment is a **decision-making tool.**
These results can be used for eco-design, environmental labeling, or strategic planning.

The objective of LCA is to present a comprehensive view of the impacts generated by products (goods, services, or processes), broken down into different simulations.



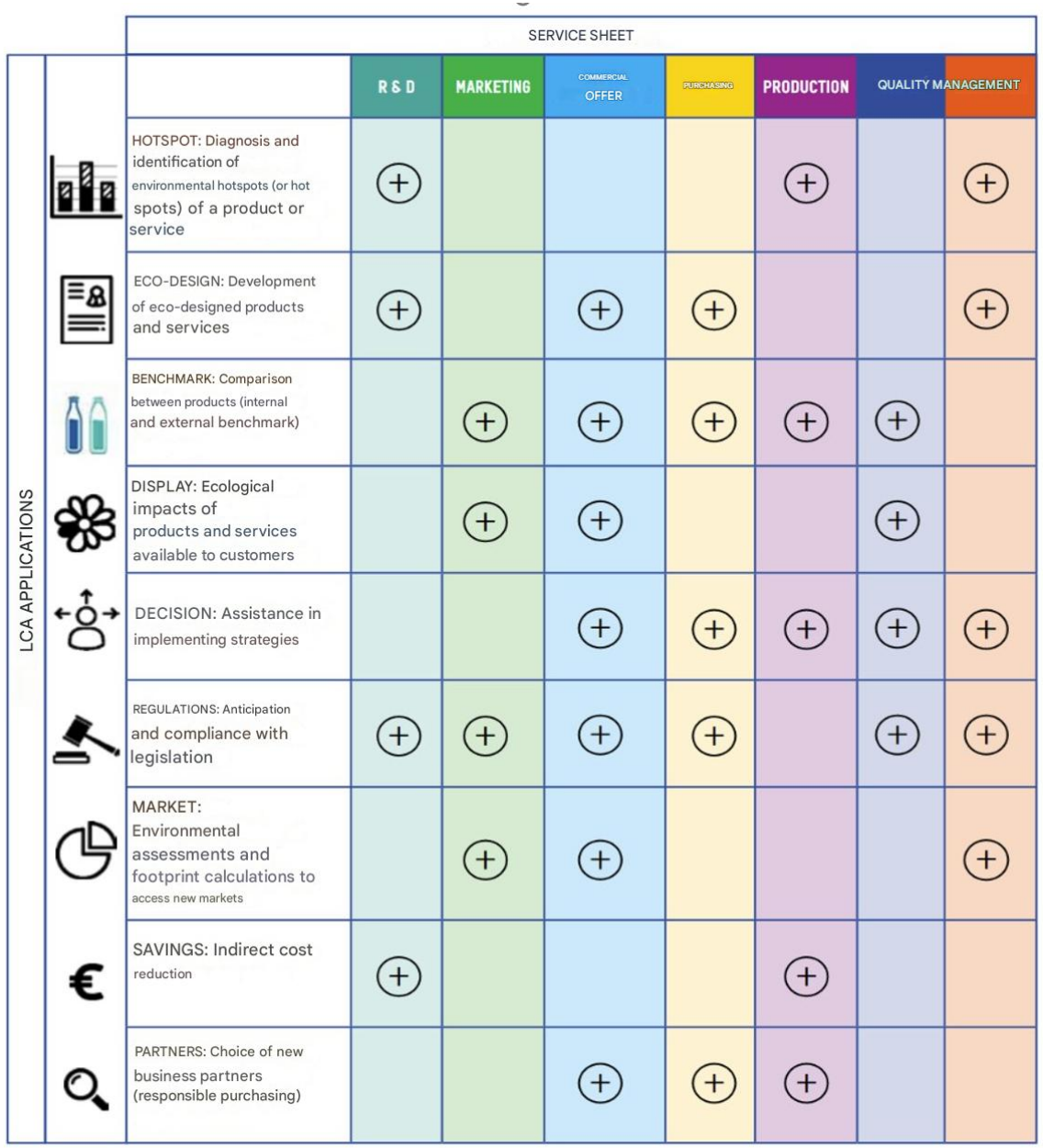
sources : Ademe

Life Cycle Analysis >> Main uses

Diagnosis
- Have a quantitative vision
- Identify priority areas to improve
- Anticipate future challenges/constraints

Compare
- 2 technical options for a designer
- 2 products (or services) for a buyer
- 2 policy directions for a decision-maker

Improve
- Implement a reduction strategy
- Establish recommendations
- Monitor changes over time



sources : Ademe, Ecoconception.fr : une activité performante et durable grâce à l'ACV

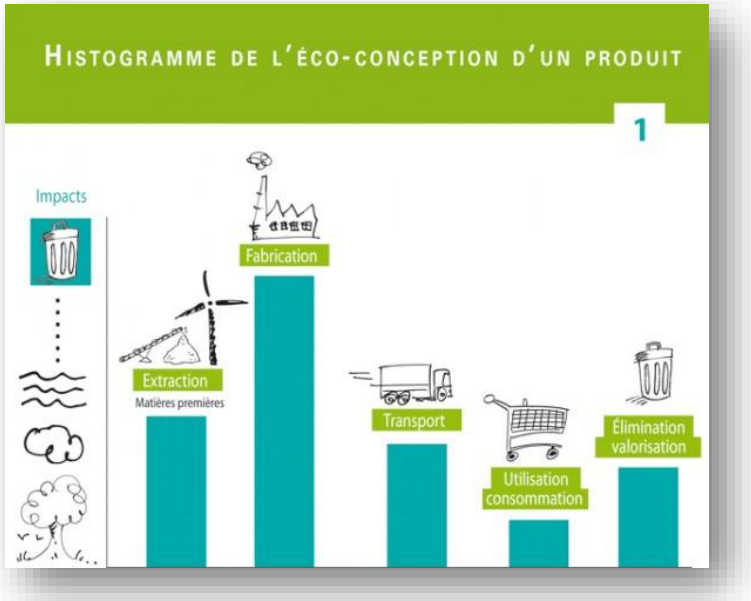
Avoid impact transfer

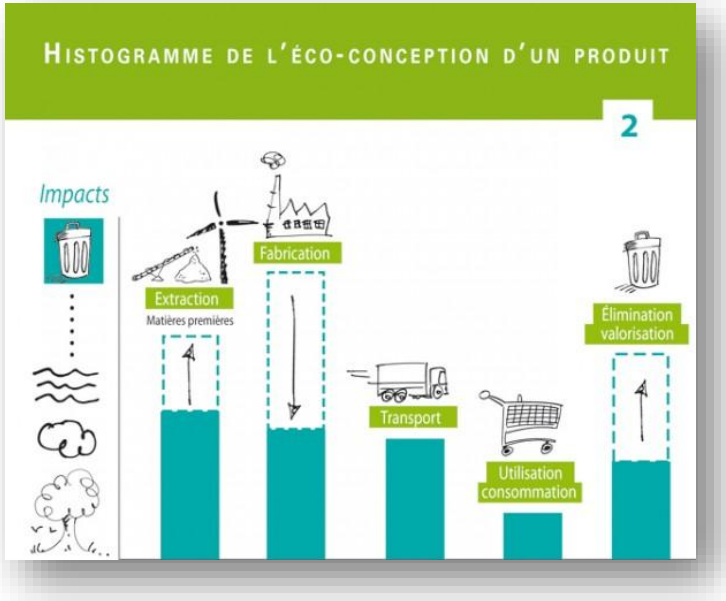
During a **comparison**, it may show that one product has less impact than another on one criterion *(greenhouse gas emissions, for example)*, but more on another criterion *(air acidification, for example).*

On the graph :

The analysis of the existing situation
(1) shows that the major environmental impact is generated during manufacturing.

The proposed solution
(2) would reduce manufacturing impact but would increase other stages (extraction and disposal): we called this, a **pollution transfer.**

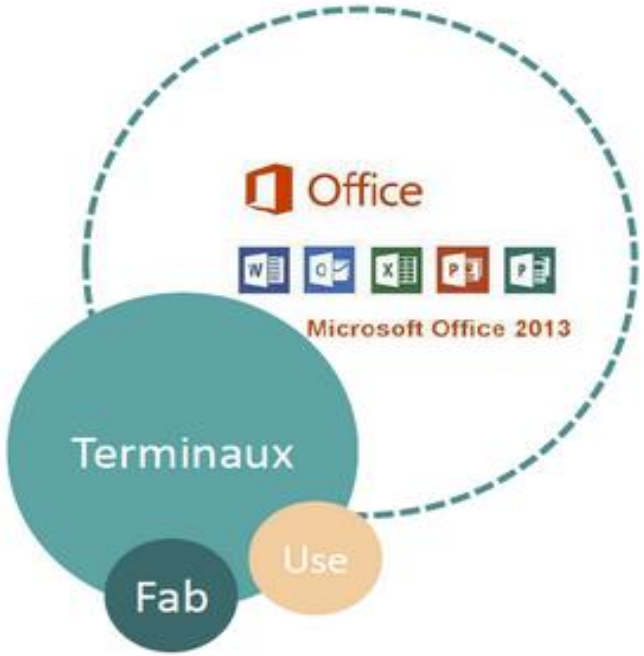


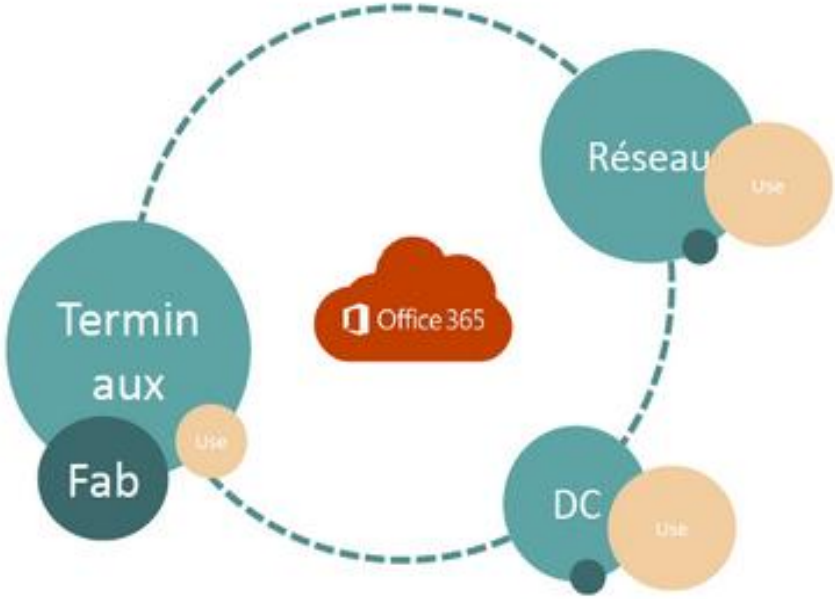


sources : Ademe, les.cahiers-developpement-durable.be

Avoid impact transfer

Software : on premise vs SaaS





SaaS mode reduces constraints on **user terminals**, particularly during the usage phase, but causes a transfer of impact to the n**etwork and data center** layers.

sources : DDemain

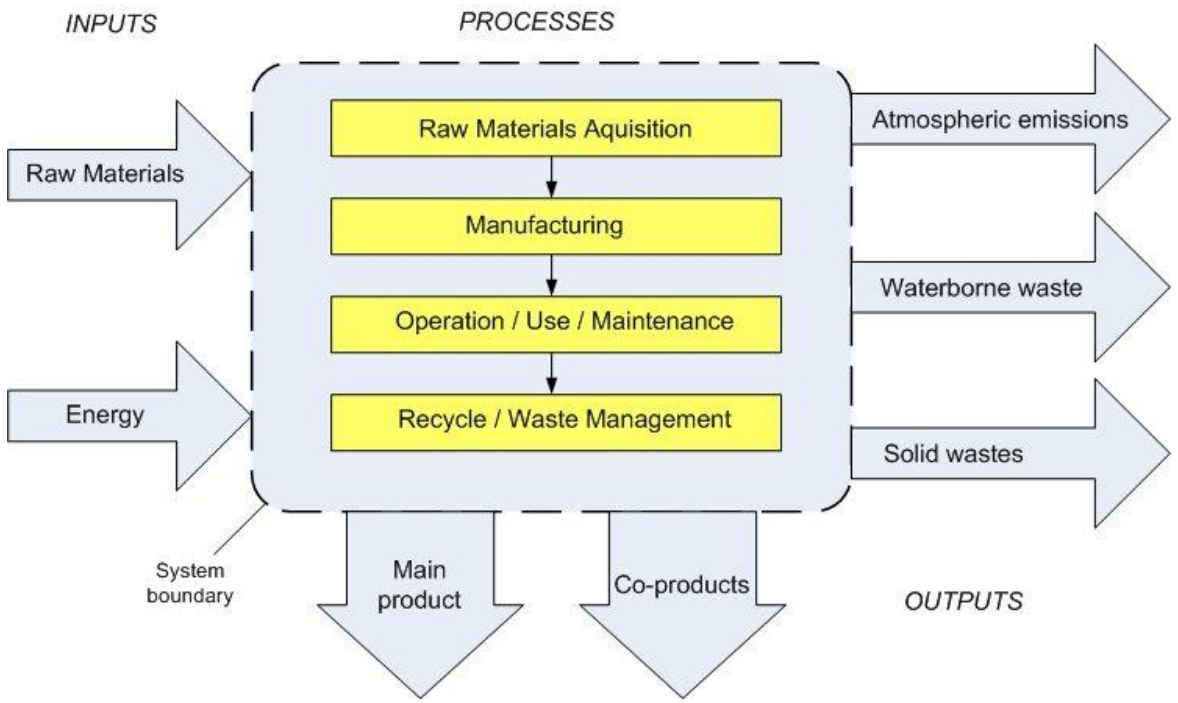
Life Cycle Analysis
>> Multi criteria

Several criteria to analyse inflows and outflows

***"flow"** : everything that goes into the manufacturing of the product and everything that comes out in terms of pollution.*

- ✓ **Inflows :** for example, those of materials and energy: resources in iron, water, oil, gas.
- ✓ **Outflows:** These may include waste, gas emissions, discharged liquids, etc.

Collecting information on flows is an **important step in LCA.**
They are quantified at each stage of the cycle and correspond to indicators of potential impacts on the environment.



sources : Ademe, https://www.e-education.psu.edu/eme807/node/690

From technical data to environmental data



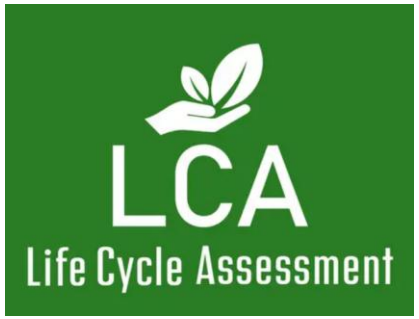
sources : DDemain

What is the difference between LCA and Carbon footprint ?



The carbon footprint consists of a single-criteria study: greenhouse gas emissions.

- Thanks to its single evaluation criterion, the carbon footprint is faster to calculate and easier to interpret.



LCA assesses a product's many potential impacts on the environment and society, based on the physical flows of matter and energy associated with it.

- LCA goes into greater depth, requiring a longer study and more work on interpreting the results.

sources : kabaun.com/post/bilan-carbone-acv, bilans-ges.ademe.fr/

Life Cycle Analysis

METHODOLOGY and practical

APPLICATION

Objectives

A product or service that performs well in a life cycle assessment is one that:

- ✓ Uses **few natural resources** (raw materials, energy, water, etc.) during its manufacture, use, and disposal.
- ✓ Uses **renewable** and/or **recycled and recyclable** natural resources during its manufacture
- ✓ Can be **reused or recycled** at the end of its life at the end of its life
- During its manufacture, use, and disposal, **generates little or no pollution**, contributes little or nothing to global warming, has little or no impact on the ozone layer, causes little or no destruction of ecosystems, and has little or no impact on biodiversity.
- ✓ During its manufacture, use, and disposal, it may **contribute to the preservation or restoration of ecosystems** and **biodiversity** or to the creation of carbon sinks.
- ✓ ...

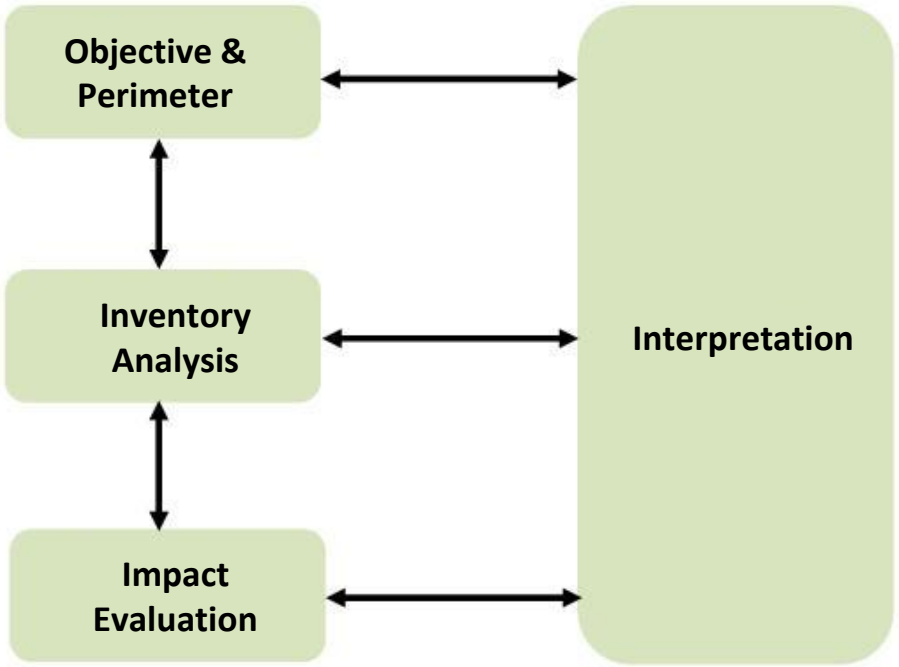
sources : les.cahiers-developpement-durable.be

Method details

According to ISO standards, four steps must be carried out during an LCA :

1. **Description of the objectives** and **scope of the study**
2. Completion and analysis of the **inventory**
3. **Impact assessment**
4. **Interpretation** of results

All these steps are interdependent and the process is iterative: changes can be made over time to ensure consistency between the different steps and refine the results.



Normalisation internationale ISO (14040 à 14044), depuis 1994.

Cadre normatif de l'étude décrit dans les normes ISO 14040 (principe et cadre) et ISO 14044 (exigences et lignes directrices).

sources : popups.uliege.be, hmf.enseeiht.fr/travaux/CD1011/bei/beiere/groupe1/node/82

APPLICATION FOR IT

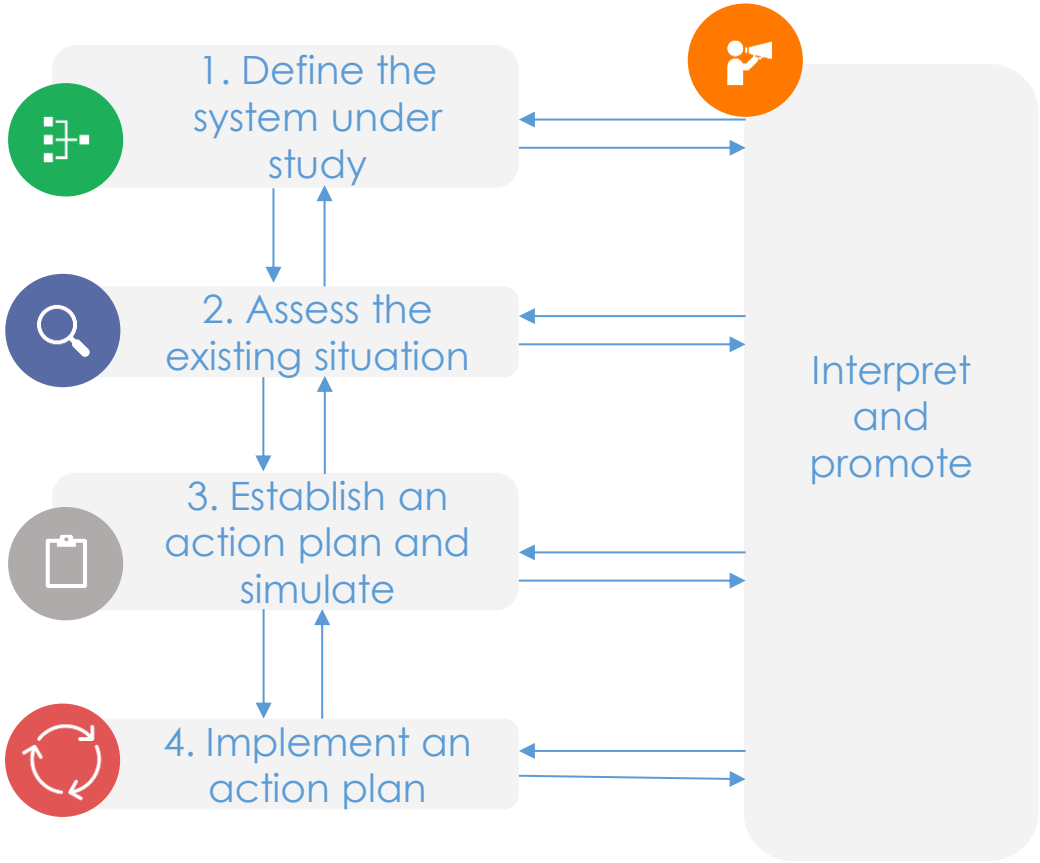
Diagnosis of an information system

- Framing workshops
- Collect data

- Analyse maturity across 74 green IT best practices
- Assess the environmental footprint (LCA)
- Analyse the beneficial and negative effects of digital technology

- Co-develop the action plan – benefit/effort
- Simulate gains

- Raise awareness among teams
- Align and integrate the action plan into the CSR and IT strategy



- Popularise indicators and communicate internally
- Environmental labelling

1 / Definition of the objectives and scope

OBJECTIVES

- ✓ Its scope: eco-design, comparison or environmental declaration, etc.
- ✓ **The target of the study and associated reasons:** market research, cost control, image risk, regulatory compliance
- ✓ **The target audience**: internal or external, novice or expert
- ✓ or if it is a **comparative statement** intended for the general public: /!\ critical review mandatory in this case



SCOPE

- ✓ The functional unit is the unit of measurement used to evaluate the service provided by the product.

sources: Ademe

1 / Definition of the functional unit

The functional unit should highlight:

- ✓ A **function** – often expressed by an action verb (deliver, transport, etc.)
- ✓ A **technical criterion** – quantifiable if possible
- ✓ A **service life**





Examples:

- For a light bulb: provide 40 W of light for 1,000 hours.
- For a smartphone: use for 2 hours per day over a period of 2 years.

Sources: ddemain.com/2018/10/31/analyse-du-cycle-de-vie-pourquoi-ne-faut-il-pas-oublier-lunite-fonctionnelle/

APPLICATION FOR IT

1 / Definition of the system to be studied

Setting the scene:

- **What are your objectives?**
- **What question do you want to answer?**



In binomial

- **Imagine a functional unit (within an IT department or a company)**

Must contain :
Function / Technical criterion (quantifiable) / Service life



APPLICATION FOR IT

1 / Definition of the system - Deliverable

Fonctionnal unit

Enable and support the association's activities through digital technology for a period of one year.

Reference flow

Digital equipment required for the IT department to operate for one year.

APPLICATION FOR IT

1 / Definition of the system - Deliverable

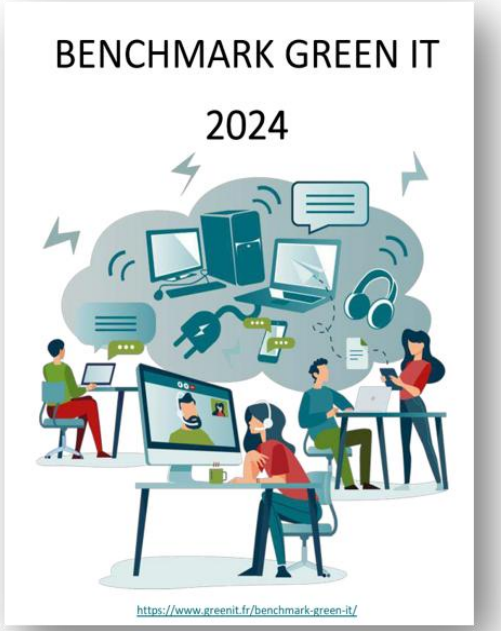
Functional Unit

"Make available and use the information system of organization XXX, by all of its users, for one year."

Information System '3 Tiers' :

- User environment
- Network
- Data Center

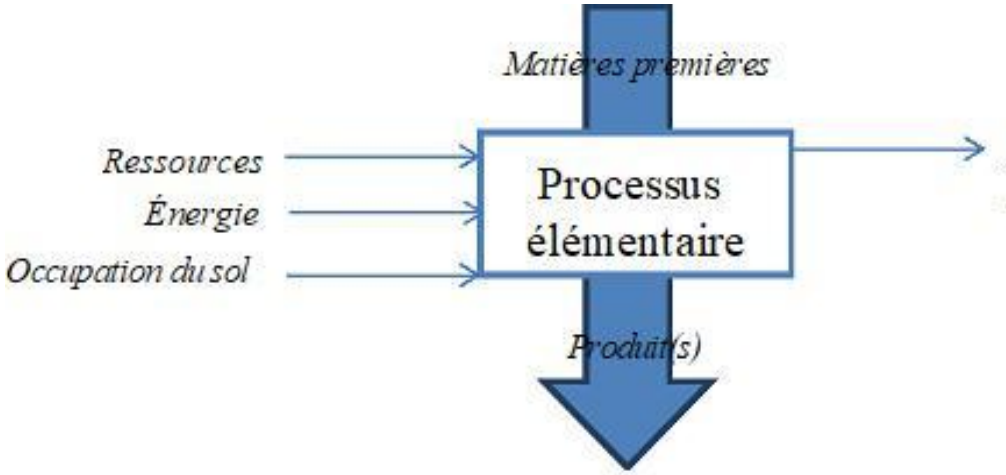
See lesson 2



2 / Inventory of life cycle - ILC

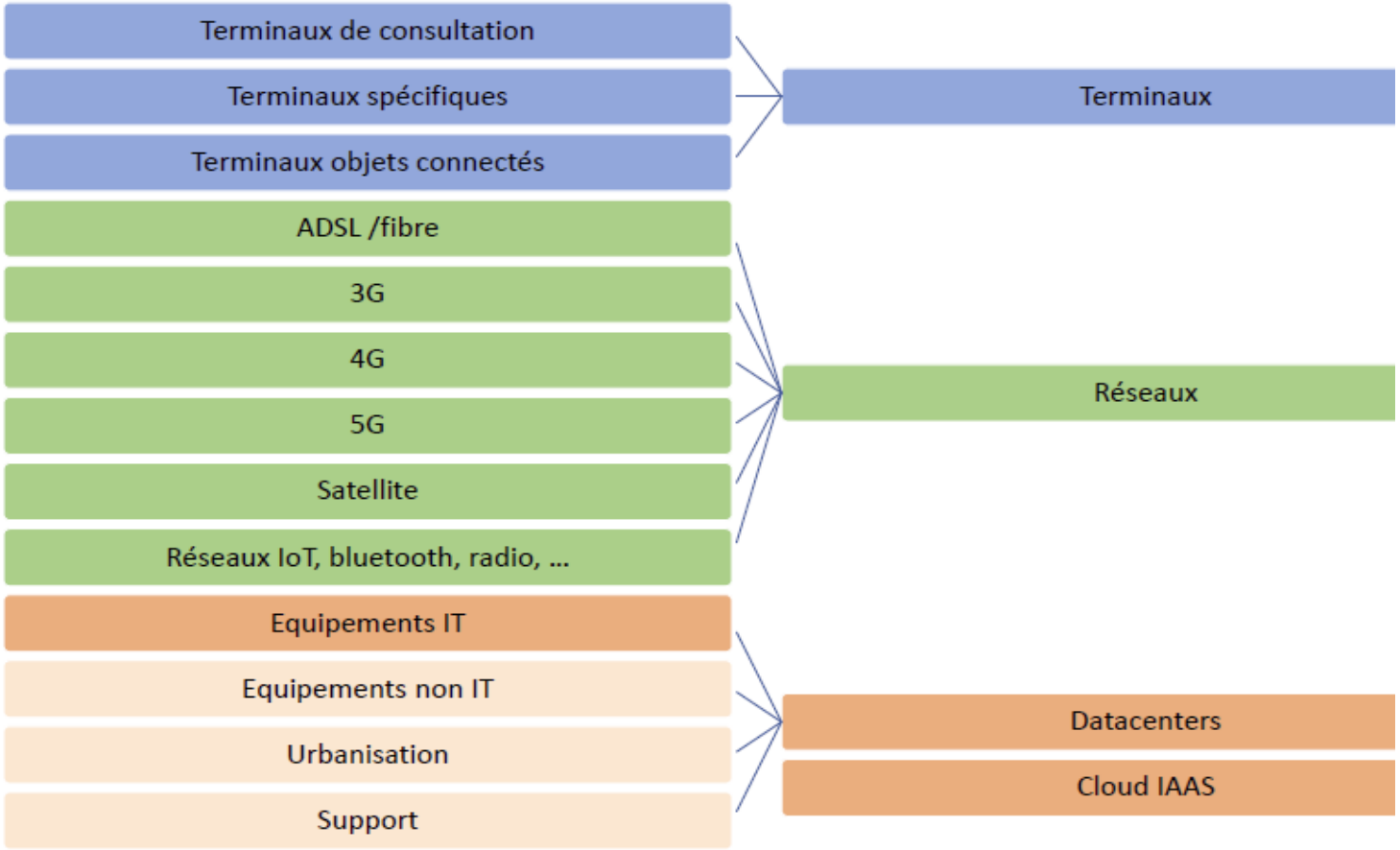
This stage involves drawing up an inventory of incoming and outgoing material and energy flows associated with the stages of the life cycle relating to the functional unit selected. The inventory is an analytical accounting of flows, which is itself broken down into two phases:

- ✓ The data **collection** phase
- ✓ The **conversion** phase **for basic flows**



Sources: Ademe

2 / Inventory : Digital service example



sources : DDemain



APPLICATION FOR IT

2 / Inventory

Environnement utilisateur

- Station d'accueil
- Écran d'ordinateur - 24 pouces, LCD
- Écran d'ordinateur - 28 pouces, LCD
- Télévision - 45 pouces, LCD
- Télévision - 53 pouces, OLED
- Télévision - 68 pouces, OLED
- Terminal libre service
- Ordinateur portable - Type bureau
- Ordinateur portable - Type Performance
- Périphériques de poste de travail
- Tablette - Milieu de gamme - 10.3 pouces
- Tablette - Haut de gamme - 11.1 pouces
- Ordinateur de bureau - Milieu de gamme
- Ordinateur de bureau - Power User

Impression

- Papier - 100% recyclé
- Papier - Fibre vierge
- Imprimante personnelle
- Imprimante multifonction

Téléphonie

- Smartphone - Milieu de gamme - 6.57 pouces
- Smartphone - Haut de gamme - 6.72 pouces
- Téléphone fixe

Réseau local (LAN) & Locaux techniques

- Serveur NAS rack - 24 To, 16 Go RAM, 2U
- DiskStation NAS - 8 To, 4 Go RAM
- Serveur rack - 1 CPU haut de gamme, 8 Go, 8 To SSD
- Switch - 24 ports
- Router - 24 ports
- Borne WiFi

2 / Inventory

APPLICATION FOR IT

Réseau étendu (WAN)

- Réseau fixe - France
- Réseau mobile - France
- Réseau fixe - Europe - Europe
- Réseau Mobile - Europe - Europe
- Pare-feu
- Switch - 24 ports
- Box internet/gateway - Fibre
- Box internet/gateway - ADSL

Centres informatiques

- Electricité - non IT
- Baie de stockage - 24 disques - QLC, 1024 Go
- Baie 42U vide
- Rack server S - 1 CPU bas de gamme, 8 Go, 1 HDD
- Rack server M - 1 CPU haut de gamme, 32 Go, 1 To SSD
- Rack server L - 2 CPU haut de gamme, 128 Go, 1 To SSD
- Rack server XL - 2 CPU haut de gamme, 256 Go, 8 To SSD
- Serveur blade (lame) - 2 CPU haut de gamme, 32 Go, 1 To SSD
- Pare-feu
- Surface totale du centre de données

Cloud

- Machine virtuelle - CH - 2 CPU, 32 Go RAM, 1 To SSD, 1 an, 1 GPU
- Machine virtuelle - CN - 2 CPU, 32 Go RAM, 1 To SSD, 1 an, 1 GPU
- Machine virtuelle - EU - 2 CPU, 32 Go RAM, 1 To SSD, 1 an, 1 GPU
- Machine virtuelle - FR - 2 CPU, 32 Go RAM, 1 To SSD, 1 an, 1 GPU
- Machine virtuelle - US - 2 CPU, 32 Go RAM, 1 To SSD, 1 an, 1 GPU
- IaaS - Stockage
- Achats de Cloud - IaaS

DSI

- Distance parcourue en avion
- Distance parcourue en train
- Distance parcourue en transport publics
- Distance parcourue en voiture
- Achats de services informatiques - Conseil et prestations
- Achats de services informatiques - Formation
- Achats de services informatiques - Maintenance
- Surface des locaux du service informatique

2 / Inventory of life cycle : European Database



The EPLCA is the EU's knowledge base that responds to business and policy needs towards sustainable production and consumption.

The EPLCA supports the methodological development of Life Cycle Assessment (LCA) for the analysis of supply chains and end-of-life waste management.

https://eplca.jrc.ec.europa.eu/index.html#menu1

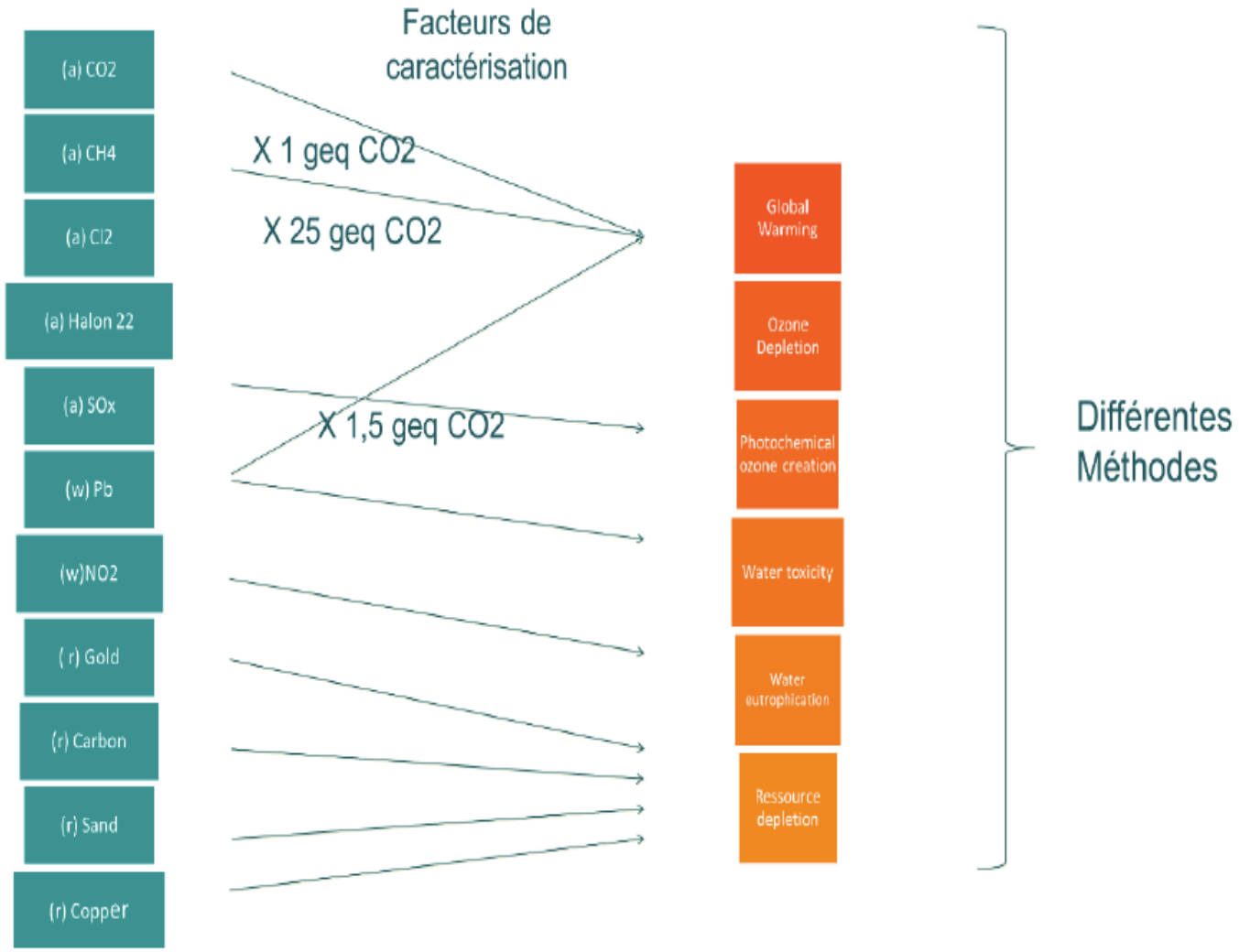
3 / Impact evaluation

There are several types of representations

- ✓ **Characterisation** allows the results of the inventory to be converted into a common unit based on their respective contributions to the impact category in question (e.g. "Climate change expressed in kg of CO2eq").

Or

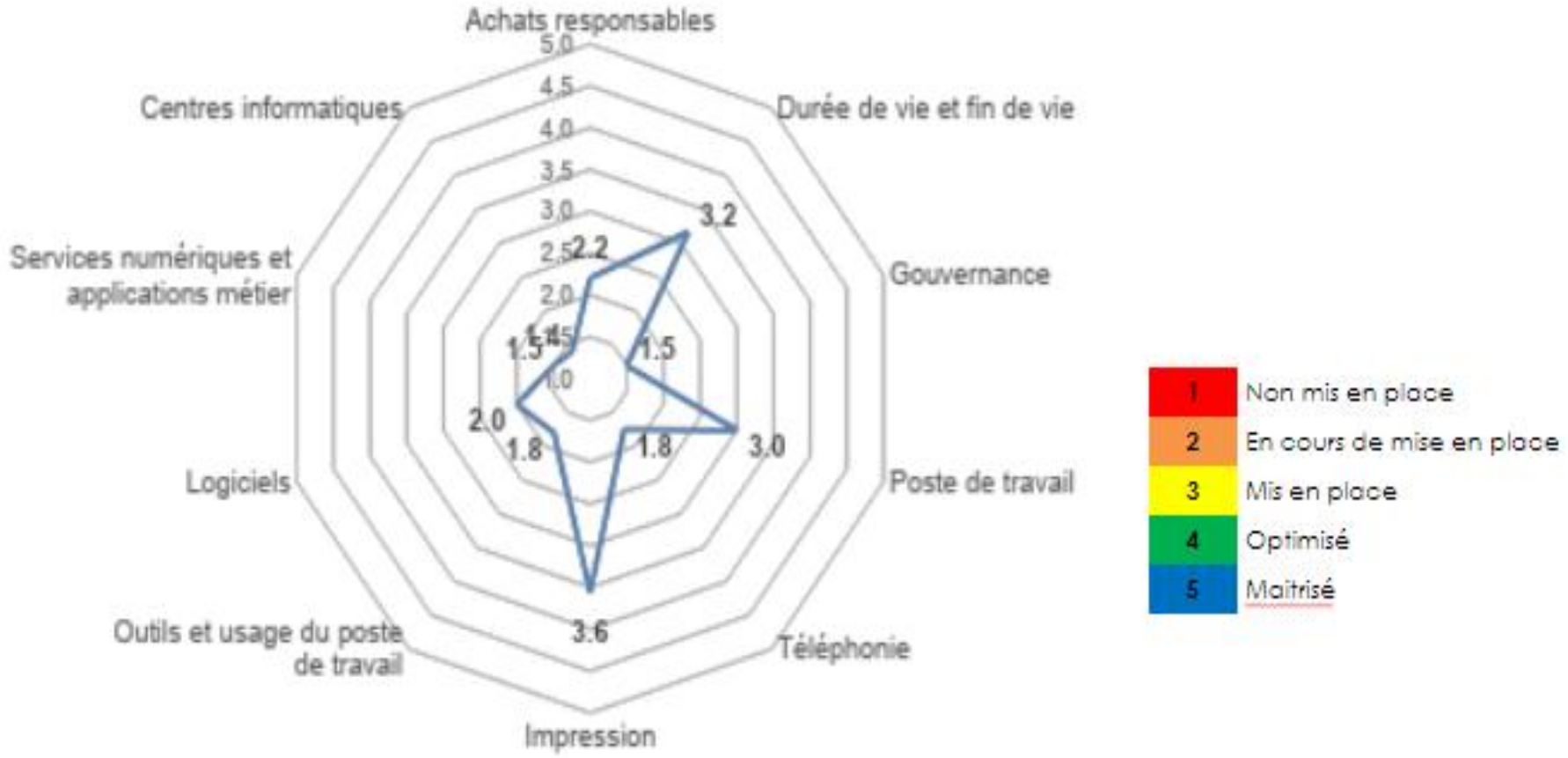
- ✓ **Normalisation**
- ✓ **Ponderation**
- ✓ **Unique score**
- ✓ **...**



Sources: hmf.enscpjht.fr/travaux/bei/beiere/book/export/html/2120_Ddomain

APPLICATION FOR IT

3/ Evaluate the existing situation - MATURITY

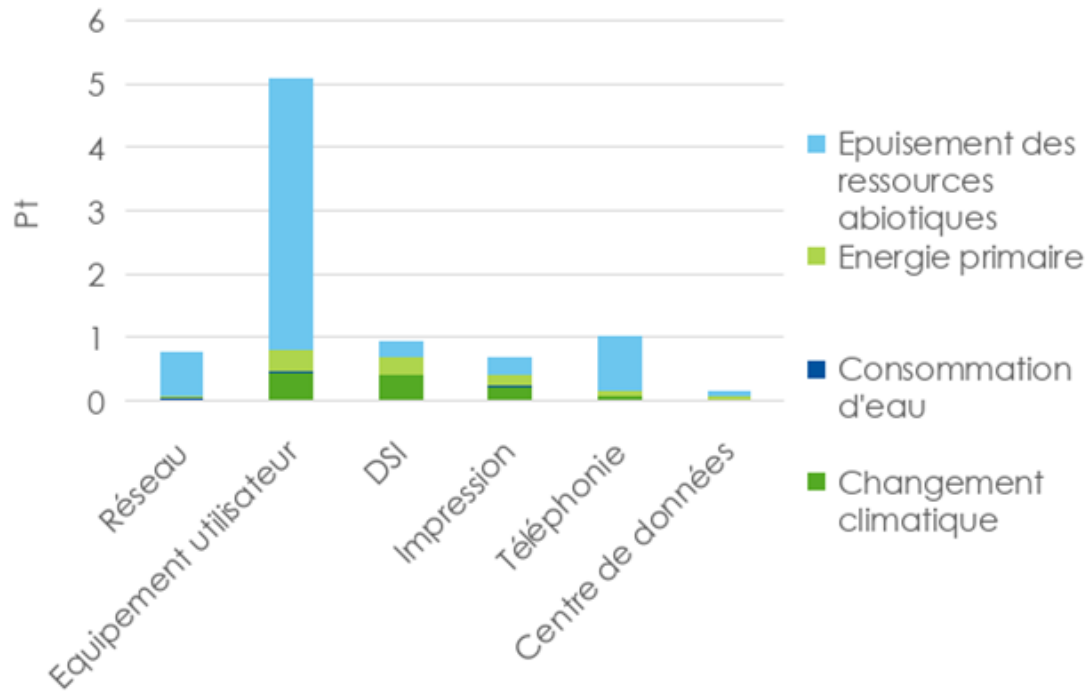


https ://club.greenit.fr/doc/2022-06-GREENIT-Referentiel_maturite-v3.pdf

APPLICATION FOR IT

3 / Evaluate the existing situation – Quantify Impact

Empreinte environnementale globale répartie par catégorie en score unique



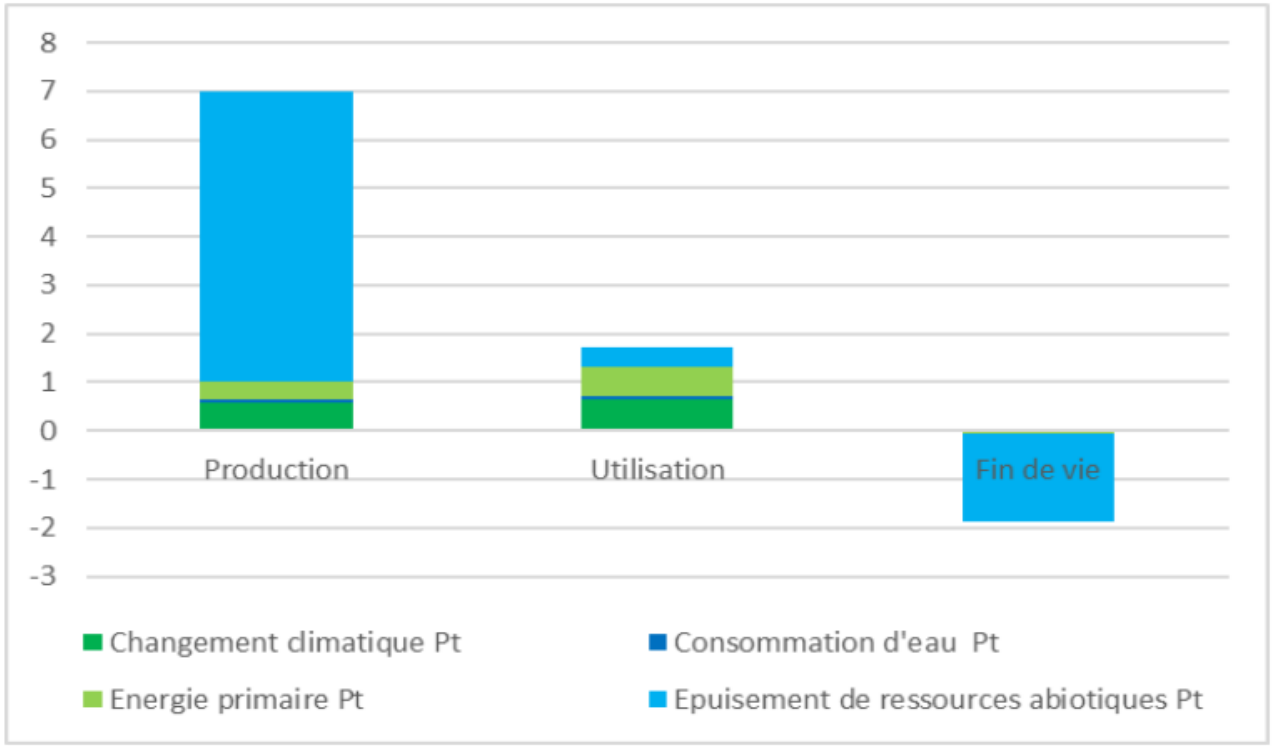
Périmètre

- 820 équipements utilisateurs,
- 312 équipements de téléphonie,
- 91 équipements réseau,
- 87 équipements d'impression,
- 14 équipements centre de données.

APPLICATION FOR IT

3 / Evaluate the existing situation – Quantify Impact

Empreinte environnementale globale répartie par étape du cycle de vie



4 / Interpretation of results

This **step is iterative** with the previous three, so as to always validate that the results obtained meet the study objectives (*for example, the unavailability of certain data may lead to restricting the scope of the study during the course of the study*).
This is also where the robustness of the results will be assessed.

The purpose of the interpretation phase is to:

- ✓ Verify the consistency of the results with the objectives and scope of the study
- ✓ Draw conclusions by explaining the limitations
- ✓ Provide recommendations

Questions to ask:

- ✓ What are the numerical results?
- ✓ Are they consistent with other known figures? A few reminders:
  - 1 average French person = 10T CO2eq/year
  - Direct emissions 1 km by car = 120 g CO2eq
  - 1 pack of water = 9L
- ✓ Is there any transfer of pollution? etc.

APPLICATION FOR IT

4 / Interpret and enhance – As an example

Numérique traditionnel



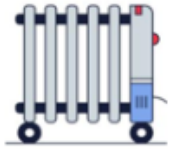
Green House Gas

- **316 667 km** en voitures
- Soit **8** tours du monde



Water

- **1 488 889** packs d'eau de 9L
- Ou **53 600** baignoires



Primary Energy

- **73,1 radiateurs** de 1000W, 8h par jour, 220 jours par an



Abiotic Resources

- **77 600 tonnes** de de terre excavée

APPLICATION FOR IT

4 / Action plan – As an example

Positive points to reinforce

- Overall lower footprint compared to the benchmark.
- WEEE is systematically entrusted to a broker (extending the life of equipment).
- Transparent partner
- Low printing volume.

Areas for improvement

- Governance: Consolidate and deploy Green IT governance in countries
- User equipment: Improve the accuracy of the monitor inventory for France. When purchasing new monitors, favour LCD technology.
- Network: Some rationalisation can be carried out between Ethernet and Wi-Fi
- IT department: Work with HR/CSR to reduce daily travel by car and plane

APPLICATION FOR IT

4 / In case of comparison – As an example

Exemple d'application de l'ACV pour comparer 2 produits :

Smartphone A has half the environmental impact of smartphone B

...but smartphone A has a battery that lasts three years, while smartphone B has a battery that lasts six years.

The impact of smartphone A must be multiplied by two in order to compare it with that of smartphone B.

>> **Result**: the actual impacts of the two smartphones are equivalent.

Option / Critical review

The purpose of the critical review is to validate a study conducted using LCA, in order to ensure that its structure complies with the rules of ISO 14044.

The critical review must guarantee that:

- ✓ The methods used to carry out the LCA are consistent with the ISO 14044 standard and valid from a scientific and technical point of view.
- ✓ The data used are appropriate and reasonable in relation to the objectives of the study.
- ✓ The interpretations reflect the identified limitations and objectives of the study.
- ✓ The study report is transparent and consistent.

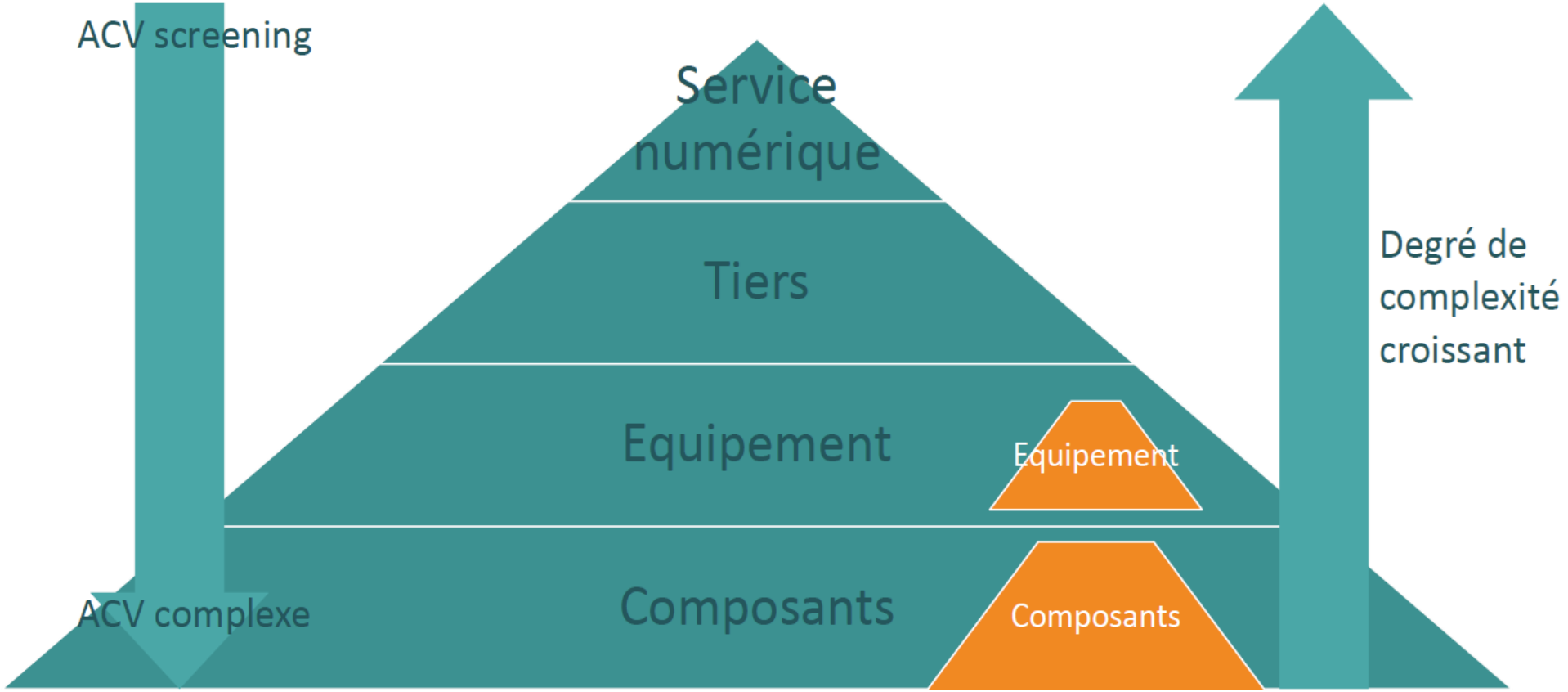
A critical review is necessary in the following cases:

- ✓ Comparison and dissemination of results
- ✓ Complex issues, research, policy guidance

Note: *A critical review cannot verify or validate the objectives set by the LCA sponsor or the use that will be made of the LCA results.*

Sources: eco-conception.fr

LCA types : Products, Services



Limitations of an LCA

According to the methodology currently in use (and despite the fact that some are measurable), aspects such as the following are not directly considered in LCAs:

- ✓ the impact of activities on landscapes,
- ✓ noise,
- ✓ odours,
- ✓ weather,
- ✓ the toxicity of emitted products (significant uncertainties).

Practical limitations may also arise: **data availability** and, above all, **data collection**, which is very cumbersome and sometimes even impossible. Results may also be influenced (intentionally or unintentionally) by the choices made by the person conducting the study in terms of assumptions or impact analysis methods.

Finally, **LCA is a tool for assessing the environmental footprint** of an industry, service or product. *It is therefore only one element of the three pillars of sustainable development, which also covers economic, social and health issues.*

Sources: stockage.univ-valenciennes.fr/MonetACVBAT20120704/acvbat/chap03/ac/ch03_040_1.4.html

Some results - Examples

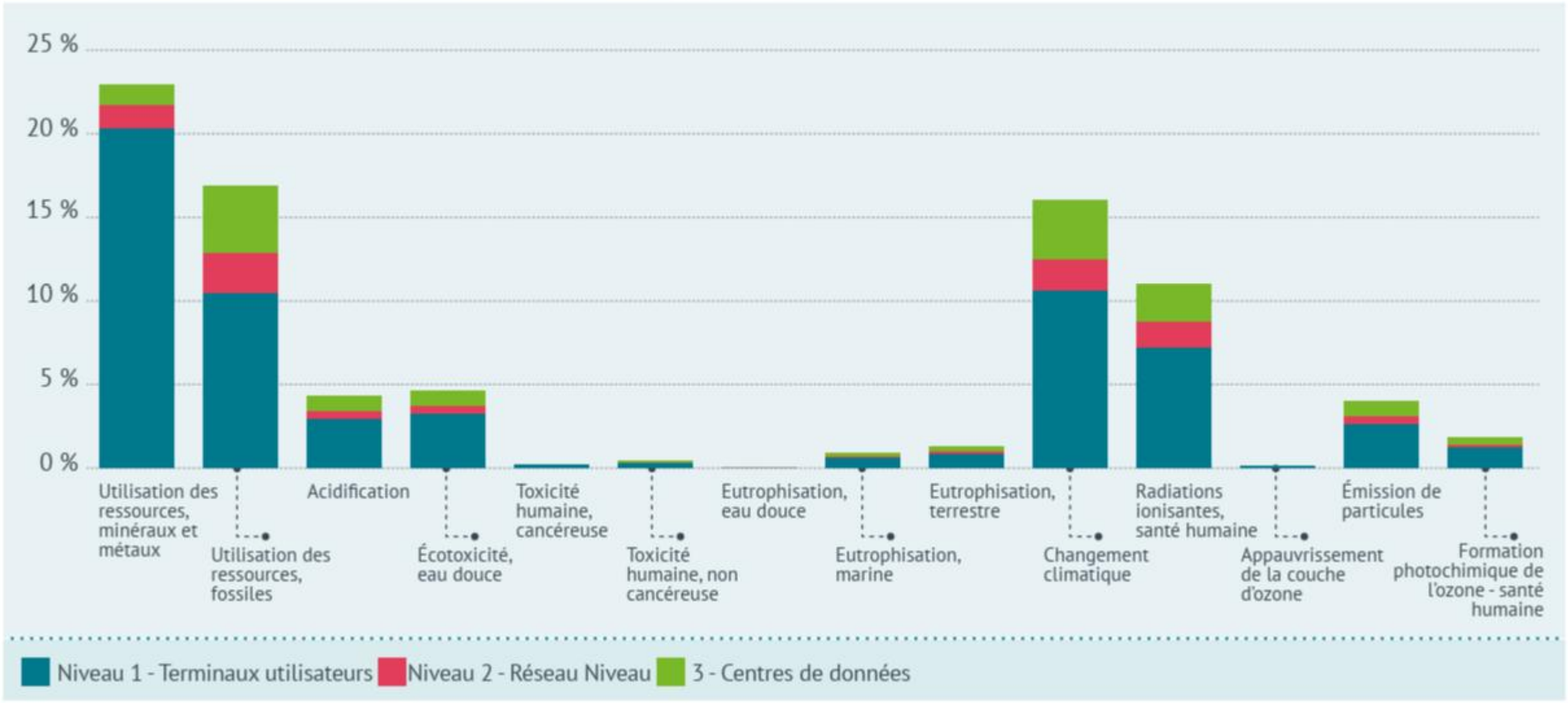
Global results of environmental impacts for one year of digital services in Europe (EU-28)

Tableau 16 - Impacts globaux des services numériques de l'UE-28 (indicateurs d'impacts environnementaux et de flux)

Utilisation des ressources, minéraux et métaux - tonnes éq. Sb	5 760
Utilisation des ressources, fossiles- PJ	3 960
Acidification - mol éq. H+ (en milliards)	1,19
Écotoxicité, eau douce - CTUe	3 090
Changement climatique - Mt éq. CO_2	185
Radiations ionisantes, santé humaine - GBq éq. U235	278
Émission de particules - Apparition de maladies	8 000
Formation d'ozone photochimique - santé humaine - tonnes éq. COVNM	464 000
Matières premières - Mt	571
Production de déchets - Mt	116
Consommation d'énergie primaire - PJ	4 230
Consommation d'énergie finale (utilisation) - PJ	1 020

sources : *Le numérique en Europe : une approche des impacts environnementaux par l'analyse du cycle de vie*

Figure 1 - Distribution normalisée et pondérée de l'impact le long des 3 niveaux



sources : *Le numérique en Europe : une approche des impacts environnementaux par l'analyse du cycle de vie*

Some results - Examples

In order to accurately **measure the impact of paper and digital media,** La Poste has chosen to invest in a benchmark study: **a Life Cycle Assessment (LCA).**

This study measures environmental impacts across 16 relevant indicators.

Automotive Brand Advertising

MEDIA ANALYZED: *A 16-pages A5 color mailing sent by addressed mail compared to a website accessible online via a link sent via promotional email.*

- Paper is more favorable than digital for 13 out of 16 environmental indicators, including climate change, fossil fuel use, human health, and acidification.
- **A paper mailing has a 1.7 times lower impact on ozone depletion than an email.**

Flyer for a restaurant chain

MEDIA ANALYZED: *A single-sided, color A5 flyer distributed through mailboxes compared to a short advertising video on social media.*

- Paper is more favorable than digital for 15 out of 16 environmental indicators, except for land use.
- **A paper flyer has a 3.3 times lower impact on global warming than a video on social media.**

https://solutionsbtob.laposte.fr/mediapositiveimpact

Act about the digital service

Lesson 6 > Ecodesign of digital service

LCA to act. How ?

Good Practices for Digital



Cross-cutting approaches
1. Responsible purchasing
2. Lifespan and end of life
3. Governance

User

4. Workstation
5. Telephony
6. Printing
7. Workstation tools and uses

Company
8. Software
9. Digital services and business applications
10. Data centres
11. Network

In groups of 4/5

Choose a user theme

- Propose 2 or 3 actions to reduce the footprint
- Specify the change (I am moving from ... to ...) and assess which environmental indicator would benefit from the impact.
- Could there be a transfer of impact?

Observation - USER



Our equipment is subject to greater obsolescence:

- Hardware
- Software
- Psychological



Observation - USER



Easy to upgrade, repair, recondition and recycle.

Difficult or impossible to upgrade, repair, recondition, recycle



1985 — 2022

average lifespan
10 to 15 years

Lifespan **divided by 3 in 30 years**

Average lifespan
4 to 6 years

Source: Sources: WWF France 2012, Guide for an eco-responsible information system and Benchmark Green IT, Green IT collective, 2014-2022 - Formatting by Academix

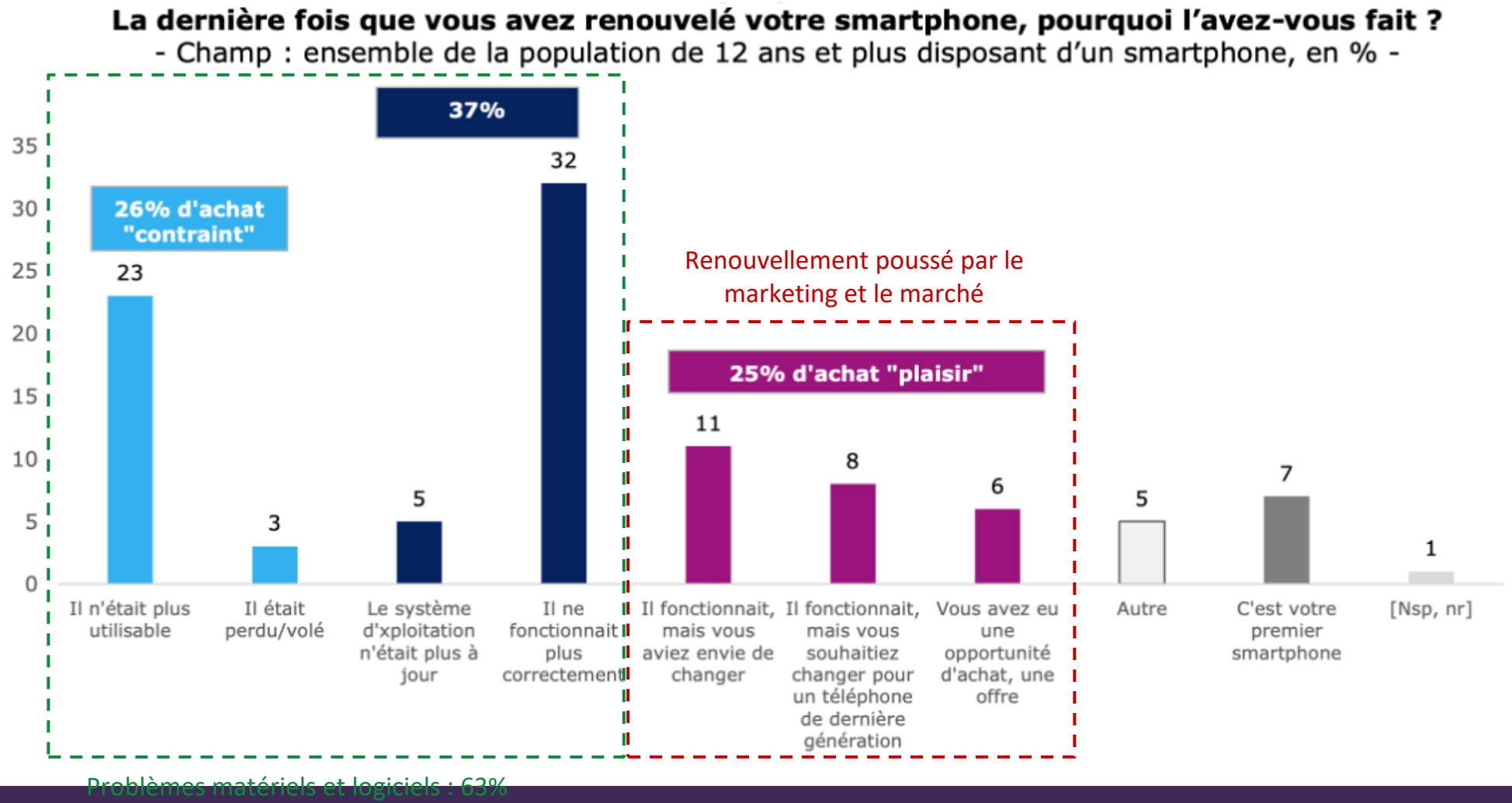
Good practices - USER

- Adjust the workstation according to need
- Upgrade equipment rather than replacing it
- Choose reconditioned equipment or, if new, equipment with an eco-label
- Optimise the power consumption of equipment
- Limit the number of screens
- Take out an extended warranty

Good Practices - USER

- Raise awareness and train employees
- Apply best practices in design to the graphic charter
- Streamline email usage

Observation - PHONE



source : ARCEP - BAROMÈTRE DU NUMÉRIQUE Édition 2021

Good Practices - PHONE

- Limit the number
- Deploy Bring Your Own Device
- Set a maximum SAR of 0.6W/kg

Observation - PRINTING

- A quarter of printouts are discarded within 5 minutes
- 16% of printouts are never read
- 65% of printouts could be read on a screen

Good Practices - PRINTING

- Raise awareness among employees
- Prioritise networked multifunction printers over local ones
- Require a badge or code to print (-30% printing)
- Adopt good printing practices (eco settings, double-sided printing, black and white, small font size)
- For printers, favour Blue Angel and Energy Star labels
- For cartridges, choose refillable ones
- For paper, choose Blue Angel or FSC labels (not PEFC)

La loi Anti Gaspillage pour une Économie Circulaire (Anti-Waste for a Circular Economy)
AGEC law - 10 février 2020



Inform consumers about the ease of repairing electrical appliances offered for sale

Raise awareness about planned obsolescence

Encourage manufacturers of electronic and electrical appliances to design **appliances that are easier to repair**

Some examples:

- Obligation to provide information on the period during which the **software updates** provided at the time of purchase remain compatible with normal use of the device.

- Obligation for government buyers to **purchase goods that are reused** or contain recycled materials.

REEN Law

"Réduire l'Empreinte Environnementale du Numérique" / "Reducing the Environmental Footprint of Digital Technology"

The law of November 15, 2021, aimed at reducing the environmental footprint of digital technology in France, known as the **REEN law**, applies to all players in the digital value chain: industry professionals, public authorities, and consumers.

1. **Raising awareness of the environmental impact of digital technology**
2. **Limiting the replacement of digital devices**
   - Strengthening the provisions of the AGEC law in favor of extending the life of products
   - Supporting reconditioning, which gives products a second life
   - Recovering "dormant stock" of unused equipment
3. **Promoting environmentally friendly digital practices**
4. **Promoting more energy-efficient data centers and networks**
   - Controlling energy consumption, improving PUE
5. **Promoting a responsible digital strategy in the regions**

Lesson 4 - As a summary

- Ecodesign
  - ISO 14001
  - Environmental evaluation
  - Multicriteria
  - Life Cycle Analysis
- LCA to choose, to compare, to decide
- Good practices users

LCA is a hard work but it is useful to choose the right product/service and avoid impact transfer

JUNIA Grande école d'ingénieurs
HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable